

## **CSE488 – Big Data Analytics**

### **Term Project**

Bangladesh National Election 2026 – Big Data Correlation & Predictive Analytics using Apache Spark

#### **1. Project Overview**

In this term project, students will analyze the results of the Bangladesh National Election 2026 and investigate the relationship between election outcomes and socio-economic factors. The project emphasizes large-scale data processing, distributed computing, machine learning, and analytical interpretation using Apache Spark and Spark MLlib.

#### **2. Project Objectives**

By completing this project, students will:

- Design and implement distributed data pipelines using Spark.
- Integrate multiple heterogeneous datasets (election + socio-economic).
- Perform correlation analysis at scale.
- Apply regression and classification models using Spark MLlib.
- Interpret analytical findings in a socio-political context.

#### **3. Technical Requirements**

Students MUST use the following technologies:

- Apache Spark (PySpark or Scala)
- Spark SQL
- Spark MLlib

#### **4. Data Sources**

Students will combine:

- Bangladesh National Election 2026 results (constituency-level)
- Bangladesh Bureau of Statistics socio-economic indicators
- World Bank / UN development indicators (if applicable)

#### **5. Data Engineering Tasks**

- Data cleaning and missing value handling
- Data normalization and feature scaling
- Joining large datasets using Spark DataFrame operations
- Feature engineering (turnout %, winning margin, urbanization index, etc.)

## **6. Analytical Components (Mandatory)**

### **6.1 Correlation Analysis (Using Spark MLlib)**

Students must compute Pearson and/or Spearman correlation between election outcomes and socio-economic indicators using Spark MLlib's Correlation module.

Example:

**Investigate the correlation between literacy rate and voter turnout.**

If the correlation coefficient  $r = 0.65$ , this indicates a moderate-to-strong positive relationship, suggesting that regions with higher literacy tend to have higher turnout.

### **6.2 Regression Analysis (Using Spark MLlib)**

Students must implement Linear Regression using Spark MLlib to predict a continuous variable.

Example:

**Predict voter turnout percentage using features such as literacy rate, poverty rate, urbanization percentage, and internet penetration.**

Model evaluation must include  $R^2$ , RMSE, and coefficient interpretation.

### **6.3 Classification Model (Using Spark MLlib)**

Students must implement at least one classification algorithm (Logistic Regression, Random Forest, or Decision Tree).

Example:

**Predict the winning political party (binary or multi-class classification) based on socio-economic indicators.**

Evaluate using accuracy, precision, recall, and F1-score.

### **6.4 Clustering (Optional but Recommended)**

**Use K-Means clustering to group constituencies by socio-economic profiles and compare the clusters' voting patterns.**